

MSAJCE — Mohamed Sathak A.J. College of Engineering

Research and Development — Complete Data Extract for RAG Model

Source URL: <https://www.msajce-edu.in/research.php> | Sections: 17 sections covered

Scraped: 2026-04-28 | Chunks: 10 | Token Target: 500 | Overlap: 100 | Model: text-embedding-3-small

1. R&D Cell — Objectives

Chunk ID: msajce_research_chunk_01 | URL: <https://www.msajce-edu.in/research.php#tab1> | ~380 tokens

Describes the purpose and objectives of the MSAJCE Research and Development Cell, including focus areas, funding agency support, and activities promoted.

Research and Development — MSAJCE

The Research and Development Cell aims to nurture research culture in the College by promoting research in newly emerging and challenging areas of Engineering, Technology, Science and Humanities.

Objectives:

1. To encourage students and faculty to carry out research in Engineering, Technology, Science and Humanities by providing necessary facilities and infrastructure.
2. To create zeal among students and faculty towards research and innovation and to encourage them to publish findings in conferences and journals.
3. To assist students to apply for funding under the student project scheme to funding agencies like TNSCST, MSME, IET (I) etc.
4. To work closely with industry to understand their needs, resulting in new or improved products, processes, systems or services, and to promote Institute-Industry Collaboration.
5. To support the government by taking up sponsored projects offered by agencies such as TNSCST, DST, AICTE, UGC, ISRO, DRDO, CSIR, DBT etc.
6. To enhance the general research capability of budding engineers by participating in conferences, seminars, workshops, and project competitions.
7. To organize workshops, seminars, and trainings related to promotion of research.
8. To encourage both students and faculty to apply for copyrights and patents.

Key funding agencies supported: TNSCST, MSME, IET (I), DST, AICTE, UGC, ISRO, DRDO, CSIR, DBT.

List of Supervisors page: <https://www.msajce-edu.in/listofsupervisors.php>

Sample Questions this chunk answers:

- What is the purpose of the MSAJCE Research and Development Cell?
- Which funding agencies does MSAJCE R&D Cell work with?
- How does MSAJCE support research among students and faculty?
- Does MSAJCE help students apply for research funding?
- What kind of research activities does MSAJCE promote?
- How does MSAJCE promote Institute-Industry Collaboration through R&D?
- Can students at MSAJCE apply for government research funding?
- What is the role of the R&D Cell at MSAJCE?

Keywords: research and development, R&D cell, MSAJCE, TNSCST, DST, AICTE, UGC, ISRO, DRDO, CSIR, DBT, MSME, IET, research funding, patents, copyrights, student project scheme, institute industry collaboration

2. Research Advisory Committee

Chunk ID: msajce_research_chunk_02 | URL: <https://www.msajce-edu.in/research.php#tab2> | ~420 tokens

Lists all members of the MSAJCE Research Advisory Committee with their designations, contact numbers, and email addresses.

Research Advisory Committee — MSAJCE

- 1 | Dr. B. Janarthanan, Head-Research, MSAJCE, Chennai | Convener
- 2 | Dr. Srinivasan, Secretary TNSCST, Chennai. Phone: 044-22301428, Email: ms.tanscst@nic.in | Advisor
- 3 | Dr. Parthiban, Asso. Prof / Prod. NIT, Trichy. Phone: 9443430246, Email: parthiban@nitt.edu | Advisor

4 | Dr. Somashekhar S Hiremath, Associate Professor, MSB 310, Manufacturing Engineering Section, Dept of Mechanical Engineering, IIT Madras, Chennai. Profile: https://iitm.irins.org/profile/61849#personal_information_panel | Advisor

5 | Dr. Sabitha Ramakrishnan, Professor, Dept of Instrumentation MIT, Chennai. Phone: 9789977989, Email: sabitha.ramakrishnan@gmail.com | Advisor

6 | All Ph.D qualified faculty | Members

Note: An alternate listing from [listofsupervisors.php](#) includes Dr. B. Sheela Rani, Dean Research, Sathyabama University, Chennai. Phone: 9840941389, Email: directorresearch@sathyabama.ac.in as Advisor (replacing Dr. Somashekhar S Hiremath in that version).

Research Advisory Committee

S.No	Name	Designation
1	Dr. B. Janarthanan, Head-Research, MSAJCE, Chennai	Convener
2	Dr. Srinivasan, Secretary TNSCST, Chennai. Ph: 044-22301428, ms.tanscst@nic.in	Advisor
3	Dr. Parthiban, Asso.Prof/Prod. NIT, Trichy. Ph: 9443430246, parthiban@nitt.edu	Advisor
4	Dr. Somashekhar S Hiremath, Associate Professor, IIT Madras, Chennai. https://iitm.irins.org/profile/61849	Advisor
5	Dr. Sabitha Ramakrishnan, Professor, Dept of Instrumentation MIT, Chennai. Ph: 9789977989, sabitha.ramakrishnan@gmail.com	Advisor
6	All Ph.D qualified faculty	Members

Sample Questions this chunk answers:

- Who is the convener of the MSAJCE Research Advisory Committee?
- Who are the advisors in the MSAJCE Research Advisory Committee?
- What is Dr. B. Janarthanan's role at MSAJCE?
- How can I contact the Research Advisory Committee at MSAJCE?
- Who from IIT Madras is an advisor to MSAJCE research?
- Who is Dr. Sabitha Ramakrishnan and what is her role at MSAJCE?
- What is the Research Advisory Committee structure at MSAJCE?
- What are the contact details for TNSCST related to MSAJCE research?

Keywords: Research Advisory Committee, Dr. B. Janarthanan, Dr. Srinivasan, Dr. Parthiban, NIT Trichy, Dr. Somashekhar Hiremath, IIT Madras, Dr. Sabitha Ramakrishnan, MIT Chennai, TNSCST, PhD faculty, convener, research advisor

[Overlap carry-forward] To encourage both students and faculty to apply for copyrights and patents. Key funding agencies supported: TNSCST, MSME, IEl (I), DST, AICTE, UGC, ISRO, DRDO, CSIR, DBT. List of Supervisors page: <https://www.msajce-edu.in/listofsupervisors.php>

3. List of Supervisors — PhD Research Supervisors

Chunk ID: [msajce_research_chunk_03](#) | URL: <https://www.msajce-edu.in/listofsupervisors.php#tab4> | ~460 tokens

Lists all MSAJCE faculty approved as PhD research supervisors under Anna University, with their reference numbers and research areas.

List of Supervisors — MSAJCE (Anna University Approved PhD Supervisors)

- 1 | Dr. K.S. Srinivasan | 1440364 | VLSI Design, Digital Signal and Image Processing
- 2 | Dr. I. Manju | 3040008 | VLSI Design, Video Processing
- 3 | Dr. B. Janarthanan | 2620054 | Machine Design, Mechanical Vibrations, Vehicle Dynamics
- 4 | Dr. G. Ramesh | 4120175 | Natural Fiber, Composite Material, Mathematical Modeling
- 5 | Dr. S. Prasath | 4220015 | Welding Technology, Materials Engineering, Manufacturing Engineering
- 6 | Dr. A. Suvitha | 2070720 | Inorganic Chemistry
- 7 | Dr. M. Vigneshkumar | 3910019 | Remote Sensing, GIS, Survey & Geology, Environmental Engineering

Full page URL: <https://www.msajce-edu.in/listofsupervisors.php>

IIC Contact Details:

Head of Institution: Dr. K.S. Srinivasan, Principal, MSAJCE, Chennai-603103. Email: principal@msajce-edu.in

IIC President: Dr. B. Janarthanan, Professor, Dept of Mechanical Engineering, MSAJCE, Chennai-603103. Email: mech.janarthanan@msajce-edu.in

IIC Vice President: Dr. I. Manju, Professor and Head, Dept of Electrical and Electronics Engineering, MSAJCE, Chennai-603103. Email: headir@msajce-edu.in

List of PhD Supervisors

S.No	Name	Anna University Ref. No	Research Area
1	Dr. K.S. Srinivasan	1440364	VLSI Design, Digital Signal and Image Processing
2	Dr. I. Manju	3040008	VLSI Design, Video Processing
3	Dr. B. Janarthanan	2620054	Machine Design, Mechanical Vibrations, Vehicle Dynamics
4	Dr. G. Ramesh	4120175	Natural Fiber, Composite Material, Mathematical Modeling
5	Dr. S. Prasath	4220015	Welding Technology, Materials Engineering, Manufacturing Engineering
6	Dr. A. Suvitha	2070720	Inorganic Chemistry
7	Dr. M. Vigneshkumar	3910019	Remote Sensing, GIS, Survey & Geology, Environmental Engineering

Sample Questions this chunk answers:

- Who are the PhD supervisors at MSAJCE?
- Which MSAJCE faculty can guide PhD research?
- What is Dr. K.S. Srinivasan's research area at MSAJCE?
- What is Dr. B. Janarthanan's Anna University reference number?
- Does MSAJCE have a supervisor for VLSI research?
- Who supervises mechanical engineering PhD at MSAJCE?
- What is the Anna University ref no for Dr. I. Manju?
- Can I do PhD in Environmental Engineering at MSAJCE?
- Who is the IIC President at MSAJCE?
- What is the email of the Principal of MSAJCE?

Keywords: PhD supervisors, Anna University, VLSI Design, Digital Signal Processing, Video Processing, Machine Design, Mechanical Vibrations, Natural Fiber, Composite Material, Welding Technology, Inorganic Chemistry, Remote Sensing, GIS, Environmental Engineering, Dr. K.S. Srinivasan, Dr. I. Manju, Dr. B. Janarthanan, Dr. G. Ramesh, Dr. S. Prasath, Dr. A. Suvitha, Dr. M. Vigneshkumar

[Overlap carry-forward] Note: An alternate listing from listofsupervisors.php includes Dr. B. Sheela Rani, Dean Research, Sathyabama University, Chennai. Phone: 9840941389, Email: directorresearch@sathyabama.ac.in as Advisor. All Ph.D qualified faculty are members of the Research Advisory Committee.

4. Publications — Faculty and Students (2016–2024)

Chunk ID: msajce_research_chunk_04 | URL: <https://www.msajce-edu.in/research.php#tab5> | ~310 tokens

Lists all available publication PDF downloads for MSAJCE faculty and students from 2016-17 to 2023-24, with direct download links.

Publications — Faculty and Students, MSAJCE

MSAJCE maintains year-wise publications records for faculty and students. The following annual publication lists are available for download:

- Publications 2023-2024: <https://www.msajce-edu.in/images/research/Publications23-24.pdf>
- Publications 2022-2023: <https://www.msajce-edu.in/images/research/Publications22-23.pdf>
- Publications 2021-2022: <https://www.msajce-edu.in/images/research/Publications21-22.pdf>
- Publications 2020-2021: <https://www.msajce-edu.in/images/research/Publications20-21.pdf>
- Publications 2019-2020: <https://www.msajce-edu.in/images/research/Publications19-20.pdf>
- Publications 2018-2019: <https://www.msajce-edu.in/images/research/Publications18-19.pdf>
- Publications 2017-2018: <https://www.msajce-edu.in/images/research/Publications17-18.pdf>
- Publications 2016-2017: <https://www.msajce-edu.in/images/research/Publications16-17.pdf>

All publication files are in PDF format and accessible directly from the Research page at <https://www.msajce-edu.in/research.php>

Sample Questions this chunk answers:

- Where can I find MSAJCE faculty publications?
- Does MSAJCE publish a list of research papers by faculty and students?
- How many years of publications does MSAJCE maintain?
- Where can I download the 2023-2024 publications list of MSAJCE?
- What is the link to MSAJCE research publications PDF?
- Does MSAJCE have a record of student research publications?
- Which journals have MSAJCE faculty published in?
- How do I access MSAJCE publication records?

Keywords: publications, faculty publications, student publications, research papers, journals, conferences, 2023-2024, 2022-2023, 2021-2022, 2020-2021, 2019-2020, 2018-2019, 2017-2018, 2016-2017, PDF download, MSAJCE research

[Overlap carry-forward] Can I do PhD in Environmental Engineering at MSAJCE? Who is the IIC President at MSAJCE? Dr. B. Janarthanan is the IIC President. Dr. I. Manju is the IIC Vice President. The list of supervisors is at <https://www.msajce-edu.in/listofsupervisors.php>

5. Patents Filed and Published

Chunk ID: msajce_research_chunk_05 | URL: <https://www.msajce-edu.in/research.php#tab6> | ~550 tokens

Complete list of patents filed and published by MSAJCE faculty members, with inventor names, patent titles, and patent numbers.

Patents Filed — MSAJCE Faculty

- 1 | Dr. E. Dhiravidachelvi | Design and implementation of a disaster management system using IoT and cloud computing techniques for a connected building to save lives with early warnings | 2020101867
- 2 | Dr. E. Dhiravidachelvi, Mrs. E. Jayanthi, Mrs. I. Suganthi, Mr. J. Raja, Mr. S. Naveenkumar | Vehicular Edge Computing Using A Smart Embedded System For Intelligent Transportation | 202041033273
- 3 | Mr. K. Vairaperumal | Battery Swapping Smart Station for Future E-Vehicle Transportation | 202141021897 A
- 4 | Mr. Chintala Venkatesh | A Vehicle Tire Control System at curved roads using intelligent interfaces and support vector machines | 202141025111 A
- 5 | Mrs. D. Hema Sumitha | Real time based IOT Enabled Smart Metering System | 202141026200 A
- 6 | Mrs. G. Premalatha, Dr. E. Dhiravidachelvi | Fundamentals of Electrical Measuring Instruments | 202141014294 A

7 | Mr. M. Kamarajan, Mr. M. Ashokkumar | Wireless sensor network for long-term environmental checking for internet of things application | 202141061248

8 | Mr. M. Kamarajan | Deep learning based automatic eye cataract detection algorithm using MATLAB | 202141061583A

9 | Mr. S.R. Mohan | Refrigerator with food warming apparatus attachment | 365189001

10 | Mr. D. Weslin, Mr. V. Vigneshwaran | A method and a device of Wireless Master Joystick Controller for Robotics | 202241071210

11 | Dr. K S Srinivasan, Mrs. Jayanthi | A Smart Device to Monitoring the Optic Cable and Inform the Users in Case of any Discrepancies | 202241071306

12 | Dr. Someshwaran, Mr. Tharani Kumar | Machine Learning-based closed-loop mixture of concrete equipment and the method | 202241071309

13 | Dr. Balakrishnan, Mr. Vinothkumar | Dedicated drone-based testing of building's strength located in a hazardous environment | 202241071308

Patent Search Reference: <https://ipindiaservices.Gov.In/Patentsearch/Patentsearch/Viewapplicationstatus>

Patents Filed

S.No	Name	Title of the Patent	Patent Number
1	Dr. E. Dhiravidachelvi	Design and implementation of a disaster management system using IoT and cloud computing techniques for a connected building to save lives with early warnings	2020101867
2	Dr. E. Dhiravidachelvi, Mrs. E. Jayanthi, Mrs. I. Suganthi, Mr. J. Raja, Mr. S. Naveenkumar	Vehicular Edge Computing Using A Smart Embedded System For Intelligent Transportation	202041033273
3	Mr. K. Vairaperumal	Battery Swapping Smart Station for Future E-Vehicle Transportation	202141021897 A
4	Mr. Chintala Venkatesh	A Vehicle Tire Control System at curved roads using intelligent interfaces and support vector machines	202141025111 A
5	Mrs. D. Hema Sumitha	Real time based IOT Enabled Smart Metering System	202141026200 A
6	Mrs. G. Premalatha, Dr. E. Dhiravidachelvi	Fundamentals of Electrical Measuring Instruments	202141014294 A
7	Mr. M. Kamarajan, Mr. M. Ashokkumar	Wireless sensor network for long-term environmental checking for IoT application	202141061248
8	Mr. M. Kamarajan	Deep learning based automatic eye cataract detection algorithm using MATLAB	202141061583A
9	Mr. S.R. Mohan	Refrigerator with food warming apparatus attachment	365189001
10	Mr. D. Weslin, Mr. V. Vigneshwaran	A method and a device of Wireless Master Joystick Controller for Robotics	202241071210
11	Dr. K S Srinivasan, Mrs. Jayanthi	A Smart Device to Monitoring the Optic Cable and Inform the Users in Case of any Discrepancies	202241071306

S.No	Name	Title of the Patent	Patent Number
12	Dr. Someshwaran, Mr. Tharani Kumar	Machine Learning-based closed-loop mixture of concrete equipment and the method	202241071309
13	Dr. Balakrishnan, Mr. Vinothkumar	Dedicated drone-based testing of building's strength located in a hazardous environment	202241071308

Sample Questions this chunk answers:

- How many patents has MSAJCE filed?
- What patents has Dr. E. Dhiravidachelvi filed?
- Does MSAJCE have patents in IoT?
- Who filed a patent for EV battery swapping at MSAJCE?
- What is the patent number for the MSAJCE disaster management IoT patent?
- Has MSAJCE filed patents in machine learning?
- Where can I verify MSAJCE patent application status?
- What patents has Mr. M. Kamarajan filed at MSAJCE?

Keywords: patent, patent filed, IoT, disaster management, vehicular edge computing, EV battery swapping, vehicle tire control, smart metering, wireless sensor network, deep learning, eye cataract detection, robotics, optic cable, machine learning, concrete, drone, Dr. E. Dhiravidachelvi, Mr. K. Vairaperumal, Mr. M. Kamarajan, patent number, IPIndia

[Overlap carry-forward] All publication files are in PDF format and accessible directly from the Research page at <https://www.msajce-edu.in/research.php>. Publications span 2016-2017 to 2023-2024.

6. Patents Published and Copyrights Filed

Chunk ID: msajce_research_chunk_06 | URL: <https://www.msajce-edu.in/research.php#tab6> | ~580 tokens

Complete list of additional patents published (beyond those filed) and all copyrights filed by MSAJCE faculty, with inventor names and reference numbers.

Patents Published (Additional entries beyond Patents Filed list) — MSAJCE

14 | Dr. B. Janarthanam | Unmanned Aerial Vehicle | 202041047806 A

15 | Mr. S. Senthil Pandi, Sankareswaran | Computer implemented method of image classification with extreme learning machine | 201941054497 A

16 | Mr. A. Syed Ismail | An AI & ML based system for building architecture & map designing | 202141050400

17 | Mrs. G. Premalatha, Dr. E. Dhiravidachelvi | Real time crowd analysis using static video surveillance | 202141014294

18 | Mr. S.R. Mohan, Mr. V. Vigneshwaran, Mr. D. Sakthivel | Study on Mechanical Characterization of Graphene reinforced Aluminium Metal Matrix Composites | 202141012155

19 | Dr. S. Prasath | Friction stir welding of dissimilar aluminium alloys using tapered threaded tool profile | 202141005812

20 | Dr. B. Muthu Kumar | Type-2 Diabetes detection and estimation from glucose levels by feature detection using machine learning algorithm | 202041023885

21 | Dr. G Ramesh, Mr. S Syed Abudhahir | Smart Painting Roller with Temperature Sensor to Detect the Temperature of Wall and Paint | 202241071203

22 | Mrs. I.S. Suganthi, Mrs. Priyadharshini | Design a Micro Strip Patch for Spectrum Utilization in Cognitive Radio Networks | 202241071307

23 | Dr. I Manju, Mr. C. Venkatesh | ML Strategy for Performance Enhancement of Phase Change Material for a Smart Control Solar Application | 202241071305

Patent Search Reference: <https://ipindiaservices.Gov.In/Patentsearch/Patentsearch/Viewapplicationstatus>

Copyright Filed — MSAJCE

1 | Dr. R. Senthilkumar, G. Ramesh | Engineering Practices Lab | L-78730/2018

2 | Mr. Chintala Venkatesh, Mrs. R. Abirami | Labview Manual | 7675/2022-CO

3 | Dr. R. Senthil Kumar, Mr. P. Saravanan, Mr. D. Sakthivel | Computer Aided Design - Course Manual | 11415/2022-CO

4 | Dr. G. Ramesh, Mr. S.R. Mohan, Mr. V. Vigneswara | Computer Aided Design And Manufacturing - Laboratory Manual | 11417/2022-CO

5 | Dr. S. Prasath, Mr. A. Muhammed Ifran, Mr. J. Rajesh | Meterology And Measurements - Laboratory Manual | 11418/2022-CO

6 | Dr. R. Senthil Kumar, Dr. S. Prasath, Mr. J. Rajesh | Manufacturing Technology - Laboratory Manual | 11420/2022-CO

7 | Dr. B. Janarthanan, Mr. P. Saravanan, Mr. S.R. Mohan | Computer Aided Simulation And Analysis - Laboratory Manual | 11412/2022-CO

8 | Dr. R. Senthil Kumar, Mr. P. Saravanan, Mr. D. Sakthivel | Computer Aided Machine Drawing - Laboratory Manual | 11409/2022-CO

9 | Dr. K.S. Srinivasan, Dr. I. Manju, Ms. K. Gayathri | Revit Architecture Lab Manual | 22218/2021-CO

10 | Dr. K. S. Srinivasan, Dr. I. Manju, Mr. K. Vairaperumal | Bot Lab Manual | 22219/2021-CO

Copyright Reference: <https://Copyright.Gov.In/Frmstatusgenuser.aspx>

Patents Published (Additional)

S.No	Name	Title of the Patent	Patent Number
14	Dr. B. Janarthanam	Unmanned Aerial Vehicle	202041047806 A
15	Mr. S. Senthil Pandi, Sankareswaran	Computer implemented method of image classification with extreme learning machine	201941054497 A
16	Mr. A. Syed Ismail	An AI & ML based system for building architecture & map designing	202141050400
17	Mrs. G. Premalatha, Dr. E. Dhiravidachelvi	Real time crowd analysis using static video surveillance	202141014294
18	Mr. S.R. Mohan, Mr. V. Vigneshwaran, Mr. D. Sakthivel	Study on Mechanical Characterization of Graphene reinforced Aluminium Metal Matrix Composites	202141012155
19	Dr. S. Prasath	Friction stir welding of dissimilar aluminium alloys using tapered threaded tool profile	202141005812
20	Dr. B. Muthu Kumar	Type-2 Diabetes detection using machine learning algorithm	202041023885
21	Dr. G Ramesh, Mr. S Syed Abudhahir	Smart Painting Roller with Temperature Sensor	202241071203
22	Mrs. I.S. Suganthi, Mrs. Priyadarshini	Design a Micro Strip Patch for Spectrum Utilization in Cognitive Radio Networks	202241071307
23	Dr. I Manju, Mr. C. Venkatesh	ML Strategy for Performance Enhancement of Phase Change Material for Smart Control Solar Application	202241071305

Copyrights Filed

S.No	Name	Title of the Copy Right	Copy Right Number
1	Dr. R. Senthilkumar, G. Ramesh	Engineering Practices Lab	L-78730/2018
2	Mr. Chintala Venkatesh, Mrs. R. Abirami	Labview Manual	7675/2022-CO

S.No	Name	Title of the Copy Right	Copy Right Number
3	Dr. R. Senthil Kumar, Mr. P. Saravanan, Mr. D. Sakthivel	Computer Aided Design - Course Manual	11415/2022-CO
4	Dr. G. Ramesh, Mr. S.R. Mohan, Mr. V. Vigneswara	Computer Aided Design And Manufacturing - Laboratory Manual	11417/2022-CO
5	Dr. S. Prasath, Mr. A. Muhammed Ifran, Mr. J. Rajesh	Meterology And Measurements - Laboratory Manual	11418/2022-CO
6	Dr. R. Senthil Kumar, Dr. S. Prasath, Mr. J. Rajesh	Manufacturing Technology - Laboratory Manual	11420/2022-CO
7	Dr. B. Janarthanan, Mr. P. Saravanan, Mr. S.R. Mohan	Computer Aided Simulation And Analysis - Laboratory Manual	11412/2022-CO
8	Dr. R. Senthil Kumar, Mr. P. Saravanan, Mr. D. Sakthivel	Computer Aided Machine Drawing - Laboratory Manual	11409/2022-CO
9	Dr. K.S. Srinivasan, Dr. I. Manju, Ms. K. Gayathri	Revit Architecture Lab Manual	22218/2021-CO
10	Dr. K. S. Srinivasan, Dr. I. Manju, Mr. K. Vairaperumal	Bot Lab Manual	22219/2021-CO

Sample Questions this chunk answers:

- How many copyrights has MSAJCE filed?
- Who filed a patent for an Unmanned Aerial Vehicle at MSAJCE?
- What is the copyright number for the Engineering Practices Lab manual at MSAJCE?
- Has MSAJCE filed patents in diabetes detection?
- What is the copyright reference website for MSAJCE?
- Who filed the copyright for the Revit Architecture Lab Manual?
- Does MSAJCE have a copyright for Computer Aided Design manual?
- What is the patent for UAV at MSAJCE and who filed it?

Keywords: patent published, copyright filed, UAV, Unmanned Aerial Vehicle, image classification, extreme learning machine, diabetes detection, friction stir welding, graphene aluminium, cognitive radio, solar application, Labview Manual, Computer Aided Design, Manufacturing Technology, Revit Architecture, Bot Lab Manual, copyright, IPR, intellectual property

[Overlap carry-forward] Patent Search Reference: <https://ipindiaservices.Gov.In/Patentsearch/Patentsearch/Viewapplicationstatus>. Dr. Balakrishnan and Mr. Vinothkumar filed patent 202241071308 for dedicated drone-based testing of building strength in hazardous environments.

7. Books and Book Chapters by MSAJCE Faculty

Chunk ID: msajce_research_chunk_07 | URL: <https://www.msajce-edu.in/research.php#tab6> | ~520 tokens

Complete list of books and book chapters authored by MSAJCE faculty members, with author names, titles, and ISBN/ISSN numbers.

Books / Book Chapters — MSAJCE Faculty

- 1 | Dr. K.S. Srinivasan | Communication Theory | 978-81-89638-43-6
- 2 | Dr. K.S. Srinivasan | Principles of Communication Engineering | 978-81-87721-95-6
- 3 | Dr. K.S. Srinivasan | Electric Circuits and Electronic Devices | 978-81-8472-085-3
- 4 | Dr. K.S. Srinivasan | Basic Electrical and Electronics Engineering | 978-81-8472-086-0
- 5 | Dr. K.S. Srinivasan | Digital Communication | 978-81-89638-19-1
- 6 | Dr. K.S. Srinivasan | Electronic Devices | 978-81-8472-254-3
- 7 | Dr. K.S. Srinivasan | Electronic Devices and Circuits | 978-81-8472-101-0
- 8 | Dr. K.S. Srinivasan | Electronic Circuits I | 978-81-8472-100-3
- 9 | Dr. K.S. Srinivasan | Electronic Circuits II | 978-81-89638-16-0
- 10 | Dr. K.S. Srinivasan | Electronics and Microprocessors | 978-81-89638-19-1

- 11 | Dr. K.S. Srinivasan | Digital Signal Processing | 81-87721-46-4
- 12 | Dr. K.S. Srinivasan | Electric Circuit Theory | 978-81-938894-2-8
- 13 | Dr. K.S. Srinivasan | Electronic Devices and Circuits (As per the JNTU Syllabus) | 978-81-89638-08-5
- 14 | Dr. E. Dhiravidachelvi | Fundamentals of Electrical Measuring Instruments | 1-978-93-83409-35-8
- 15 | Dr. E. Dhiravidachelvi | Data Science for COVID-19 (Elsevier) | 978-0-12-824536-1
- 16 | Dr. E. Dhiravidachelvi | Applications of Big Data in Health Care | 978-0-820203-6
- 17 | Dr. E. Dhiravidachelvi | Monograph on Fundamentals of electrical measuring instruments | 978-93-83409-35-8
- 18 | Dr. K.S. Srinivasan, Dr. P. Subramanian, Mr. D. Weslin | Wireless Sensor Networks | 978-81-954927-4-9
- 19 | Mr. C. Daniel | Impact of Lockdown for Research Scholars | 978-93-90853-50-2
- 20 | Mr. S. Senthil Pandi | Machine Learning | 978-93-91373-85-6
- 21 | Mr. S. Senthil Pandi | Computer Networks | 202041033273
- 22 | Dr. I. Manju, Dr. E. Dhiravidachelvi, Mrs. G. Premalatha | Digital Communication and Computer Networks | 819549276-2
- 23 | Dr. K.S. Srinivasan | Electronics and Microprocessors | 978-81-89638-19-1
- 24 | Dr. K.S. Srinivasan | Digital Signal Processing | 81-87721-46-4

Books and Book Chapters

S.N o	Name	Title	ISBN/ISSN/Number
1	Dr. K.S. Srinivasan	Communication Theory	978-81-89638-43-6
2	Dr. K.S. Srinivasan	Principles of Communication Engineering	978-81-87721-95-6
3	Dr. K.S. Srinivasan	Electric Circuits and Electronic Devices	978-81-8472-085-3
4	Dr. K.S. Srinivasan	Basic Electrical and Electronics Engineering	978-81-8472-086-0
5	Dr. K.S. Srinivasan	Digital Communication	978-81-89638-19-1
6	Dr. K.S. Srinivasan	Electronic Devices	978-81-8472-254-3
7	Dr. K.S. Srinivasan	Electronic Devices and Circuits	978-81-8472-101-0
8	Dr. K.S. Srinivasan	Electronic Circuits I	978-81-8472-100-3
9	Dr. K.S. Srinivasan	Electronic Circuits II	978-81-89638-16-0
10	Dr. K.S. Srinivasan	Electronics and Microprocessors	978-81-89638-19-1
11	Dr. K.S. Srinivasan	Digital Signal Processing	81-87721-46-4
12	Dr. K.S. Srinivasan	Electric Circuit Theory	978-81-938894-2-8
13	Dr. K.S. Srinivasan	Electronic Devices and Circuits (JNTU)	978-81-89638-08-5
14	Dr. E. Dhiravidachelvi	Fundamentals of Electrical Measuring Instruments	1-978-93-83409-35-8
15	Dr. E. Dhiravidachelvi	Data Science for COVID-19 (Elsevier)	978-0-12-824536-1
16	Dr. E. Dhiravidachelvi	Applications of Big Data in Health Care	978-0-820203-6
17	Dr. E. Dhiravidachelvi	Monograph on Fundamentals of electrical measuring instruments	978-93-83409-35-8

S.No	Name	Title	ISBN/ISSN/Number
18	Dr. K.S. Srinivasan, Dr. P. Subramanian, Mr. D. Weslin	Wireless Sensor Networks	978-81-954927-4-9
19	Mr. C. Daniel	Impact of Lockdown for Research Scholars	978-93-90853-50-2
20	Mr. S. Senthil Pandi	Machine Learning	978-93-91373-85-6
21	Mr. S. Senthil Pandi	Computer Networks	202041033273
22	Dr. I. Manju, Dr. E. Dhiravidachelvi, Mrs. G. Premalatha	Digital Communication and Computer Networks	819549276-2

Sample Questions this chunk answers:

- How many books has Dr. K.S. Srinivasan authored?
- Has MSAJCE faculty published books with Elsevier?
- What is the ISBN of the MSAJCE book on Wireless Sensor Networks?
- Who authored the book on Data Science for COVID-19 at MSAJCE?
- What books has Dr. E. Dhiravidachelvi written?
- Is there a textbook on Digital Signal Processing by MSAJCE faculty?
- What is the ISBN for the book on Machine Learning by MSAJCE?
- Has MSAJCE faculty authored books on healthcare and big data?

Keywords: books, book chapters, ISBN, Dr. K.S. Srinivasan, Dr. E. Dhiravidachelvi, Communication Theory, Digital Signal Processing, Wireless Sensor Networks, Data Science COVID-19, Elsevier, Big Data Healthcare, Machine Learning, Computer Networks, Electronic Circuits, VLSI, textbook

[Overlap carry-forward] Copyright Reference: <https://Copyright.Gov.In/Frmstatusgenuser.aspx>. MSAJCE has filed 10 copyrights including Engineering Practices Lab (L-78730/2018) and multiple laboratory manuals for Computer Aided Design, Manufacturing Technology, and Revit Architecture.

8. National Innovation and Start-up Policy (NISP) — Vision, Mission, Objectives and Committee

Chunk ID: msajce_research_chunk_08 | URL: <https://www.msajce-edu.in/research.php> | ~510 tokens

Details the MSAJCE National Innovation and Start-up Policy including vision, mission, short/long-term objectives, and the NISP committee members with designations.

National Innovation and Start-up Policy (NISP) — MSAJCE

Vision:

To cater to the needs of student entrepreneurs with innovative ideas, thereby introducing a culture of entrepreneurship inside campus which will strengthen the education system and promote national economic and social growth.

Mission:

To develop an ecosystem with required infrastructure that can enable students and faculty to innovate and prototype their potential ideas with industrial standards and support from Government, industry and reputed academic institutions around the world, and help them to realize their potentials.

Short Term Objectives:

1. To help student groups to prototype their ideas.
2. To improve innovation, creative and design thinking among student community.
3. Incubation facility for faculty-driven start-up and student/Alumni start-up.
4. Organize FDP, seminars, workshops, and distinguished talks for students, Faculty and Alumni to promote entrepreneurial culture.
5. Strengthen institute-industry interaction cell activity and effectively use outcomes for achieving the mission.

Long Term Objectives:

1. Associate with DST, CII, MSME and other academic institutions for transferring world-class facility for MSAJCE IEDC activities.

2. Improve quality of research work among students and attain patents that can be commercially used in production.
3. Provide a platform for students to develop innovative products with global recognition and generate business opportunities.
4. Generate revenues through consultancy work and student start-ups.
5. Spread awareness to students and faculty regarding IPR related activities.
6. Strategic partnership linkage with:
 - a. Entrepreneurship Development Institute of India (EDII), Chennai.
 - b. National Innovation Foundation (NIF) of India to submit ideas and apply for schemes.
 - c. Procure fund from AICTE for Entrepreneurship Development Cell.

NISP Committee Members:

- 1 | Dr. K.S. Srinivasan, Principal, MSAJCE, Chennai | President
- 2 | Mr. Prakadesh Subramanian, Strategic Partner, SIIF, Chennai | Incubation Coordinator
- 3 | Mr. Parvez Aalam, CEO, Crescent Innovation Incubation Council, Chennai | Incubation Coordinator
- 4 | Mr. Thiyagaraja Gupta, Deputy Controller of Patents and Designs | Patent Expert
- 5 | Ahamed Jamel AMT, IT Consultant, 4A Technology | Alumni Entrepreneur
- 6 | Abdur Rahim Salih, Director & CEO V5 Innovations Pvt. Ltd | Alumni Entrepreneur
- 7 | Asim Ali. L, CMO, Customer Labs | Alumni Entrepreneur

NISP Committee Members

S.No	Name	Position
1	Dr. K.S. Srinivasan, Principal, MSAJCE, Chennai	President
2	Mr. Prakadesh Subramanian, Strategic Partner, SIIF, Chennai	Incubation Coordinator
3	Mr. Parvez Aalam, CEO, Crescent Innovation Incubation Council, Chennai	Incubation Coordinator
4	Mr. Thiyagaraja Gupta, Deputy Controller of Patents and Designs	Patent Expert
5	Ahamed Jamel AMT, IT Consultant, 4A Technology	Alumni Entrepreneur
6	Abdur Rahim Salih, Director & CEO V5 Innovations Pvt. Ltd	Alumni Entrepreneur
7	Asim Ali. L, CMO, Customer Labs	Alumni Entrepreneur

Sample Questions this chunk answers:

- What is the NISP policy at MSAJCE?
- Who is the president of the NISP committee at MSAJCE?
- Does MSAJCE have an incubation facility for student start-ups?
- What are the long-term objectives of MSAJCE NISP?
- Who is the patent expert in the MSAJCE NISP committee?
- Does MSAJCE have partnerships for entrepreneurship development?
- What is the vision of MSAJCE for innovation and start-ups?
- Who are the alumni entrepreneurs in the MSAJCE NISP committee?
- Does MSAJCE support student start-ups through EDII or NIF?
- What is IEDC at MSAJCE?

Keywords: NISP, National Innovation and Start-up Policy, entrepreneurship, incubation, start-up, IEDC, IPR, DST, CII, MSME, EDII, NIF, Entrepreneurship Development Cell, AICTE, FDP, innovation, prototype, Dr. K.S. Srinivasan, Mr. Parvez Aalam, alumni entrepreneur

[Overlap carry-forward] MSAJCE has authored 24 books and book chapters. Dr. K.S. Srinivasan has authored 13+ books covering Communication Theory, Digital Signal Processing, Electronic Circuits. Dr. E. Dhiravidachelvi has authored books including Data Science for COVID-19 (Elsevier, ISBN: 978-0-12-824536-1).

9. NISP Thrust Areas, KPI Monitoring and 5-Year Activity Plan

Details the NISP strategic thrust areas, key performance indicators for monitoring innovation and entrepreneurship outcomes, and the tentative 5-year activity plan.

NISP Thrust Areas — MSAJCE

- 1 | To develop Strategies and governance for Promoting Innovation Entrepreneurship in the institute
- 2 | Creating Innovation Pipeline and Pathways for Entrepreneurs and Startups
- 3 | Building Organizational Capacity, Human Resources and Incentive
- 4 | Collaboration, Co-creation, Business Relationship and Knowledge Exchange
- 5 | Incubation & Pre-Incubation support at SIIF
- 6 | IP Ownership Rights for Technologies Developed at MSAJCE and SIIF
- 7 | Pedagogy & Learning Interventions for Supporting Innovations & Start-ups
- 8 | Entrepreneurial Performance Impact Assessment

KPI Monitor & Evaluation:

Hierarchy | Key Performance Indicators (KPIs) | Means and Verification

Vision | Increase in Self-Employment Rate; 5 Established Start-ups per year | NIRF Rankings

Goal | Establish environment with multiple levels of support for Innovation, Entrepreneurship & Start-ups; Students will choose Entrepreneurship as career | ARIIA Rankings

Outcomes | 5 IPR/Innovations developed per year; 5 Student/Early-Stage Start-ups to be formed per year; 25% of student & faculty mass exposed to awareness programs; 250 beneficiaries accessing infrastructure per day/month/year; 5% student projects turned to innovations; 5 IPR-based products generated per year; 5 Research Studies on Entrepreneurship published; 2 Regional, National and International linkages established; 5 Beneficiaries referred to incubators/investors | Quarterly Newsletter; Biannual Survey; Monthly progress report

Activities | 5 Education/Skill certification programs on Entrepreneurship/IPR/Innovation; 25 workshops/awareness/market outreach events; 12 networking events per year; 5 skill/FDP/EDP programs; 1 National Hackathon organized; 1% of total budget/year spent on start-up mandate | Biannual Survey; Quarterly Newsletter; Monthly progress report

Tentative Plan for the Next 5 Years (selected activities):

- Entrepreneurship and Innovation as Career Opportunity Workshop — 1 per year
- Problem Solving/Design Thinking/Ideation/Campus Hackathon Workshop — 1 per year
- Field/Exposure Visit to Village/Society/School/Industry — 1 per year
- Special Talk on My Story - Entrepreneur's Life — 2 per year
- National Conference/Workshop on Start-up/Social Innovation & Entrepreneurship — 1 per year
- Demo Day – Exhibition Cum Demo for PoCs & Mentorship Session — 2 per year
- Internship at Innovation & Start-up Centre/Startups during Semester Break — 1 per year
- Business Plan Contest — 1 per year
- One Day Awareness/Mentoring on IPR & IP Management — 1 per year
- Boot camp for Innovation product development — 1 per year
- Innovation Day Celebrations — 1 per year
- National Science Day — 1 per year
- CITBIF Hackathon — 1 per year
- Short Term Training course on Innovation/Start-up & Entrepreneurship — 1 per year

NISP KPI Monitor and Evaluation

Hierarchy	Key Performance Indicators (KPIs)	Means and Verification
Vision	5 Established Start-ups per year; Increase in Self-Employment Rate	NIRF Rankings
Goal	Multiple level support for innovation; Students choose Entrepreneurship as career	ARIIA Rankings

Hierarchy	Key Performance Indicators (KPIs)	Means and Verification
Outcomes	5 IPR/Innovations/year; 5 Start-ups/year; 25% student & faculty exposed to awareness; 250 beneficiaries/day; 5% student projects commercialized	Quarterly Newsletter; Biannual Survey; Monthly progress report
Activities	5 certification programs; 25 workshops/year; 12 networking events/year; 1 National Hackathon; 1% budget for start-ups	Biannual Survey; Quarterly Newsletter; Monthly progress report

Sample Questions this chunk answers:

- What are the KPIs for MSAJCE innovation and start-up policy?
- How many start-ups does MSAJCE aim to establish per year?
- What is the 5-year plan for innovation activities at MSAJCE?
- Does MSAJCE organize hackathons?
- What is the CITBIF Hackathon at MSAJCE?
- Does MSAJCE conduct business plan contests?
- How does MSAJCE measure entrepreneurship outcomes?
- What percentage of MSAJCE student projects are targeted to be commercialized?
- What events does MSAJCE organize for start-ups and entrepreneurs?
- Does MSAJCE have a Demo Day for student innovations?

Keywords: NISP thrust areas, KPI, start-up targets, 5-year plan, hackathon, CITBIF Hackathon, business plan contest, innovation day, National Science Day, Demo Day, PoC, incubation, SIIF, ARIIA, NIRF, IPR management, EDP, FDP, Design Thinking, entrepreneurship workshop

[Overlap carry-forward] NISP Long Term Objective: Strategic partnership with EDII Chennai, NIF of India, and AICTE Entrepreneurship Development Cell. NISP Committee President: Dr. K.S. Srinivasan. Patent Expert: Mr. Thiagaraja Gupta, Deputy Controller of Patents and Designs.

10. Institution Innovation Council (IIC-MSAJCE) — About, Members and Events

Chunk ID: msajce_research_chunk_10 | URL: <https://www.msajce-edu.in/listofsupervisors.php> | ~580 tokens

Describes the Institution Innovation Council (IIC-MSAJCE) constituted under MHRD, its focus areas, council members with designations, and the full list of IIC events conducted.

Institution Innovation Council (IIC-MSAJCE)

IIC-MSAJCE was constituted as per the guidelines of the Ministry of Human Resource Development (MHRD) Innovation Cell (MIC).

Introduction:

MHRD, Govt. of India established 'MHRD's Innovation Cell (MIC)' to systematically foster the culture of Innovation amongst all Higher Education Institutions (HEIs). MIC's primary mandate is to encourage, inspire and nurture young students by supporting them to work with new ideas and transform them into prototypes during their formative years.

MIC has envisioned creating 'Institution's Innovation Council (IICs)' across selected HEIs to promote a local innovation ecosystem.

Major Focus of IIC:

- Create a vibrant local innovation ecosystem
- Start-up supporting mechanism in HEIs
- Prepare institute for Atal Ranking of Institutions on Innovation Achievements (ARIIA)
- Establish functional ecosystem for scouting ideas and pre-incubation of ideas
- Develop better Cognitive Ability for Technology Students

Functions of IICs:

- Identify and reward innovations and share success stories
- Organize periodic workshops/seminars/interactions with entrepreneurs, investors and professionals
- Network with peers and national entrepreneurship development organizations
- Create an Institution's Innovation portal to highlight innovative projects

- Organize Hackathons, idea competitions, mini-challenges with industry involvement

IIC Council Members:

- 1 | Dr. K. S. Srinivasan, Principal | Head of the Institution
- 2 | Dr. B. Janarthanan, Professor, MSAJCE | President
- 3 | Dr. I. Manju, Professor, MSAJCE | Vice President
- 4 | Dr. M. Kanthababu, Professor, Anna University | External Member
- 5 | Dr. N. Alagumurthi, Professor, Pondicherry Engineering College | External Member
- 6 | Mr. N. S. Srinivasan, DGM, L&T Infrastructure Development Projects Ltd | External Member
- 7 | Venkatakrishnan Sankaran, Vice-President, Levergent Technologies | External Member
- 8 | Mr. D. Gowri Shankar, Asst. Professor, New Prince Shri Bhanavani College of Engineering | External Member from Mentor Institute
- 9 | Dr. R. Senthilkumar, Associate Professor | Member
- 10 | Dr. G. Ramesh, Associate Professor | Member
- 11 | Dr. A. Balakrishnan, Associate Professor | Member
- 12 | Dr. G. Premalatha, Associate Professor | Member
- 13 | Dr. B. Devibala, Assistant Professor | Member

IIC Contact:

Head of Institution: Dr. K.S. Srinivasan, Principal, MSAJCE, Chennai-603103. Email: principal@msajce-edu.in

IIC President: Dr. B. Janarthanan, Dept of Mechanical Engineering, MSAJCE. Email: mech.janarthanan@msajce-edu.in

IIC Vice President: Dr. I. Manju, Professor & Head, Dept of EEE, MSAJCE. Email: headir@msajce-edu.in

Selected IIC Events:

1. Design of Flexible Pavement using IRC — Self-Driven Activity
2. Virtual Workshop on Machine Learning — Self-Driven Activity
3. Online Workshop on Programming Skills — Self-Driven Activity
4. Technological Innovation and Academia-Industry Collaboration under Self-Reliant India — Self-Driven Activity
5. FDP on Kinematics of Machinery — Self-Driven Activity
6. FDP on Teaching Engineering Graphics using ICT Tools — Self-Driven Activity
7. Virtual Workshop on Research Methodologies — Self-Driven Activity
8. National Webinar on Research, Innovation and Ranking — MIC Driven Activity
9. Orientation session for all students & faculties by Innovation Ambassador — IIC Calendar Activity
10. Session on How to plan for Start-up and Legal & Ethical Steps — IIC Calendar Activity
11. Workshop on Design Thinking, Critical Thinking and Innovation Design — IIC Calendar Activity
12. Workshop on Entrepreneurship and Innovation as Career Opportunity — IIC Calendar Activity
13. Awareness Programme on IPR — Self-Driven Activity
14. Session on Problem Solving and Ideation Workshop — IIC Calendar Activity
15. Internal Hackathon to select team for SIH-2022 — Self-Driven Activity
16. Workshop on Entrepreneurship Skill, Attitude and Behaviour Development — IIC Calendar Activity
17. Alumni Talk and Interaction — Self-Driven Activity
18. AKAM — MIC Driven Activity
19. IIC Annual Report — MIC Driven Activity

IIC Council Members

S.No	Name	Position
1	Dr. K. S. Srinivasan, Principal, MSAJCE	Head of the Institution
2	Dr. B. Janarthanan, Professor, MSAJCE	President

S.No	Name	Position
3	Dr. I. Manju, Professor, MSAJCE	Vice President
4	Dr. M. Kanthababu, Professor, Anna University	External Member
5	Dr. N. Alagumurthi, Professor, Pondicherry Engineering College	External Member
6	Mr. N. S. Srinivasan, DGM, L&T Infrastructure Development Projects Ltd	External Member
7	Venkatakrishnan Sankaran, Vice-President, Levergent Technologies	External Member
8	Mr. D. Gowri Shankar, Asst. Professor, New Prince Shri Bhanavani College of Engineering	External Member from Mentor Institute
9	Dr. R. Senthilkumar, Associate Professor	Member
10	Dr. G. Ramesh, Associate Professor	Member
11	Dr. A. Balakrishnan, Associate Professor	Member
12	Dr. G. Premalatha, Associate Professor	Member
13	Dr. B. Devibala, Assistant Professor	Member

Sample Questions this chunk answers:

- What is the IIC at MSAJCE?
- Who is the IIC President at MSAJCE?
- What events does the IIC-MSAJCE organize?
- Who are the external members of the IIC at MSAJCE?
- Does MSAJCE participate in the Smart India Hackathon?
- What is the role of MHRD Innovation Cell at MSAJCE?
- Who is the IIC Vice President at MSAJCE?
- What is ARIIA and how is MSAJCE involved?
- Does MSAJCE have an innovation portal?
- What workshops has the IIC at MSAJCE conducted?

Keywords: IIC, Institution Innovation Council, MHRD, MIC, Innovation Cell, ARIIA, hackathon, Smart India Hackathon, SIH-2022, IPR awareness, Design Thinking, entrepreneurship, Dr. B. Janarthanan, Dr. I. Manju, Anna University, Pondicherry Engineering College, L&T, Levergent Technologies, innovation ecosystem, start-up support

[Overlap carry-forward] NISP 5-year plan targets 5 start-ups per year, 5 IPR innovations per year, 12 networking events per year, Business Plan Contest (1/year), Innovation Day Celebrations (1/year), CITBIF Hackathon (1/year). 1% of total institution budget/year to be spent on start-up mandate.

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